SARDER JUNAID AHMED

PhD Applicant in Computational Science
Machine Learning Researcher — Methodological Rigor

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Portfolio • GitHub • LinkedIn
IELTS: Overall Band 7.0 (no band less than 6.5) • Bangladeshi

Research Objective: To replace ad hoc thresholds and unvalidated assumptions with statistically rigorous, causally grounded frameworks for governance, environmental, and policy analytics.

EDUCATION

Bachelor of Science in Computer Science & Engineering **Feb 2025**
Rajshahi University of Engineering & Technology (RUET), Bangladesh
CGPA: 3.08/4.00 • Thesis Grade: A+ • Seminar Grade: A
Research-intensive undergraduate experience; strong quantitative and methodological training; thesis awarded highest distinction

Higher Secondary Certificate (HSC) **2018**
Pabna Cadet College, Bangladesh • **Perfect GPA: 5.00/5.00**
Form Leader (led 60+ cadets); represented college at Spelling Bee, advancing to Divisional Round on three occasions

Secondary School Certificate (SSC) **2016**
Pabna Cadet College, Bangladesh • **Perfect GPA: 5.00/5.00**

RESEARCH INTERESTS

Computational Science; Statistical Validity in Machine Learning; Causal Inference; Governance Analytics; Measure-Theoretic Probability; Assumption Validation in Predictive Modeling; Development Economics; Robust Statistical Inference; Mathematical Methods in Data Science.

PUBLICATIONS

Presented Papers:

- **Ahmed, S.J.** (2025). "Multi-Dimensional Statistical Similarity for Governance Classification: Beyond Arbitrary Thresholds in Comparative Politics." *6th Annual Paper Meet Electrical Engineering Division (APMEE 2025).* Oral Presentation. **Received Best Research Paper Award.**
- **Ahmed, S.J.**, Islam Nahian, M.T., & Kwoshik, M.H.R. (2026). "CF-EGAT: A Causal Fairness-Aware Equity Graph Attention Network for Country-Level Environmental Livability Classification." *Symposium on Photonics, Emerging Computational Technologies, Research & AI-Data Science (SPECTRA 2026)*, IEEE Photonics Society HSTU SBC. Track: Artificial Intelligence & Data Science. Oral Presentation. **Received 1st Best Paper Award.** *(Proceedings DOI forthcoming)*
- **Ahmed, S.J.**, Islam Nahian, M.T., & Kwoshik, M.H.R. (2026). "RMA-BO: Regret-Minimizing Adaptive Bayesian Optimization." *Symposium on Photonics, Emerging Computational Technologies, Research & AI-Data Science (SPECTRA 2026)*, IEEE Photonics Society HSTU SBC. Track: Artificial Intelligence & Data Science. Oral Presentation. *(Proceedings DOI forthcoming)*

Manuscripts Under Review (First-Author):

- **Ahmed, S.J.**, Islam Nahian, M.T., & Kwoshik, M.H.R. (2025). *Machine Learning for Crime Classification: A Fairness-Aware Approach to Class Imbalance.*
- **Ahmed, S.J.**, Islam Nahian, M.T., Kwoshik, M.H.R., & Nakib, F.N. (2025). *Environmental Livability Assessment via Adaptive Bootstrap-Retrained SHAP and Statistically-Constrained Pareto Counterfactuals: A Cross-National Analysis.*

RESEARCH EXPERIENCE

Researcher 2026 – Present
Royal Scientific Publications, Bangladesh

- Conduct rigorous mathematical research and develop novel solutions addressing open problems in applied mathematics and computational science
- Collaborate on peer-reviewed publications synthesizing mathematical theory with practical applications

Senior Researcher & Executive Committee Member May 2022 – May 2024
Young Learners' Research Lab, Rajshahi University of Engineering & Technology (RUET)

- Conducted ML analysis, statistical inference, and numerical analysis for lab-wide initiatives; led hypothesis testing (Mann-Whitney U, Kolmogorov-Smirnov, Shapiro-Wilk) with Benjamini-Hochberg FDR correction across cross-disciplinary datasets
- Mentored 5+ junior researchers on Bayesian and causal inference; one mentee published their first conference paper as a direct outcome
- Established reproducible research standards and open science protocols adopted lab-wide
- *Awards:* Esteemed Alumni Award (2024); Certificate of Appreciation for Research Mentorship Excellence (2024)

Independent Research Projects 2021 – Present
Department of Computer Science & Engineering, RUET

- *Annual Enterprise Survey EDA (2026):* One-Way ANOVA ($p < 0.001$) across all categorical variables on 55,620 enterprise records spanning 117 industries (New Zealand); hypothesis-driven statistical framework with professional visualizations; GitHub
- *Political Regime Classification:* Classified governance across 127 nations with 94% accuracy using hypothesis testing, FDR correction, and bootstrap validation; 95.4% consistency across 100 randomized configurations
- *Economic Development Classification:* 98.3% accuracy classifying 150+ countries on World Bank indicators via principled feature selection; GitHub
- *Crime Classification & Public Safety:* 93% accuracy on 6-class crime prediction (XGBoost) with spatial-temporal features and distributional assumption validation; GitHub

PROJECTS

StatsPro — AI-Powered Statistical Analysis Platform 2025 – 2026
Live App • Deployed on Streamlit Cloud

- Deployed a full-stack platform automating CSV-to-report workflows with 10 statistical tests (Shapiro-Wilk, ANOVA, chi-square, Mann-Whitney U, OLS, VIF) and Benjamini-Hochberg FDR correction throughout
- Automated preprocessing (imputation, outlier capping), 15+ visualizations (Q-Q plots, heatmaps, pairplots), and instant data quality scoring
- Built ML Studio: 6 models (XGBoost, Random Forest, CatBoost) with 5-fold cross-validation, live inference, and LLM-generated executive insights

IMDb Transformer Sentiment Analysis — DistilBERT Fine-Tuning 2025
GitHub

- Fine-tuned pretrained **DistilBERT** on 10,000 IMDb reviews; **87.05% test accuracy**, loss reduced 0.4449 → 0.1615 over 3 epochs (AdamW, linear warmup, T4 GPU)
- **Technical:** HuggingFace Transformers, PyTorch; benchmarked against classical ML baseline to quantify Transformer trade-offs on limited data

IMDb Sentiment Analysis — NLP & Classical ML Comparison 2025
GitHub

- Classified 50,000 IMDb reviews via TF-IDF; Logistic Regression achieved **89.76% accuracy**, 90.9% recall (stratified 80/20 split), outperforming LinearSVC and Naive Bayes
- **Technical:** Unigram + bigram TF-IDF (10k features), scikit-learn; pipeline serialized as production-ready `.pkl` artifacts

TEACHING EXPERIENCE

Independent Tutor & Peer Mentor 2019 – Present
Self-employed, Dhaka, Bangladesh

- Provided free or low-cost tutoring in Physics, Chemistry, Higher Mathematics, ICT, and English to 50+ students from economically disadvantaged backgrounds; also mentored 10+ university peers in Vector Calculus, Statistics, Linear Algebra, and Machine Learning
- Mentored 5 students to achieve **perfect GPAs (5.00/5.00)** on national SSC and HSC board examinations
- Guided 4 students to gain admission to premier engineering universities (RUET, BUET, CUET)

HONORS AND AWARDS

2026	1st Best Paper Award, SPECTRA 2026 (CF-EGAT)
2025	Best Research Paper Award, APMEE 2025 Conference
2024	Esteemed Alumni Award, Young Learners' Research Lab, RUET
2024	Certificate of Appreciation for Research Mentorship Excellence, RUET
2018	Perfect GPA (5.00/5.00), Higher Secondary Certificate (HSC)
2016	Perfect GPA (5.00/5.00), Secondary School Certificate (SSC)
2013, 09	Talentpool Scholarship (National Merit-based)
2016	General Scholarship (National Merit-based)
2013	Principal's Prize for Outstanding Academic Performance
2016–2018	Best Speller Award (×2), Pabna Cadet College

TECHNICAL SKILLS

Programming:	Python, C/C++, SQL, Git, Docker, Linux, LaTeX
Statistical Methods:	Probability Theory, Bayesian Inference, Causal Inference, Hypothesis Testing (Mann-Whitney U, Kolmogorov-Smirnov, Shapiro-Wilk), FDR Correction, Bootstrap
Machine Learning:	TensorFlow, Scikit-learn, XGBoost, Random Forest, Ensemble Methods, Feature Engineering, Cross-validation
Deep Learning:	TensorFlow, PyTorch, HuggingFace Transformers, DistilBERT Fine-Tuning, Neural Network Optimization
NLP:	TF-IDF Vectorization, Sentiment Analysis, Transformer Fine-Tuning
Mathematical Methods:	Numerical Analysis, Differential Equations, Linear Algebra, Optimization Theory
Data Visualization:	Matplotlib, Seaborn, Plotly
Research Tools:	Jupyter, Git/GitHub, Linux, LaTeX

COMMUNICATION & CREATIVE PURSUITS

Radio Jockey & Show Host 2020–2021
Radio Kothabondhu, Dhaka, Bangladesh

Creative Works

- **Published Poet:** Author of poetry collection "Sob Odvuture" (Bengali)
- **Lyricist, Composer & Singer:** Write lyrics, compose, and perform original songs.
- **Poetry Recitation:** First place in multiple competitions (2010–2013).

REFERENCES

Md. Nahiduzzaman
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Thesis supervisor; can speak to research independence and methodological rigor

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Research mentor; can speak to technical growth and publication contributions